

Maximum likelihood estimation

Pick the parameters that make the observed data most probable, then ask how far off they are on average.

THE QUESTION

Given data and a model family, which parameter values should we report, and how good are they?

THE INTUITION

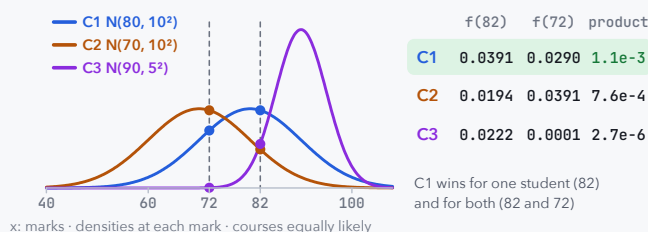
Fix the data and read the density as a function of the parameters: that is the likelihood. Independent points multiply, so take logs and add, then set the gradient to zero.

KEY TAKEAWAY

For a normal, the MLE is the sample mean and the $1/n$ covariance. The mean is unbiased; the $1/n$ variance is low by a factor $(n-1)/n$.

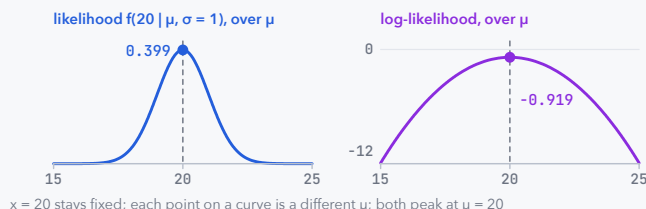
model $p(x | \theta) \rightarrow$ i.i.d. data $D \rightarrow$ likelihood: product $\rightarrow$ log-likelihood: sum $\rightarrow$ gradient = 0 $\rightarrow$ MLE $\rightarrow$ check its bias

1 Pick the course that makes the marks most probable



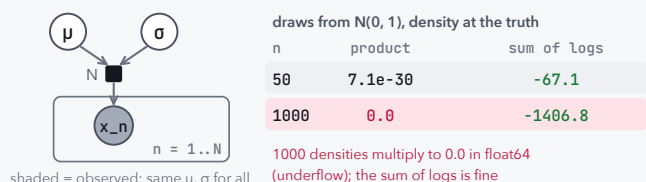
Marks in three courses are normal. A mark of 82 has the highest density under C1. A second student from the same course says 72; the product of the two densities still favours C1. This compares $p(D | \text{course})$ only; unequal course sizes need priors (see the MAP and priors sheet).

2 Fix the data, vary the parameter



One value $x = 20$ from $\mathcal{N}(\mu, 1)$: $\mathcal{L}(\mu) = f(20 | \mu, 1)$ is a function of μ with x fixed. Log is increasing, so $\log \mathcal{L}$ peaks at the same μ . $\mathcal{L}$ is not a density over θ : for the coins D1 of panel 4, $\int_0^1 \mathcal{L}(\theta) d\theta = 1/495$.

3 Independent draws multiply, so their logs add

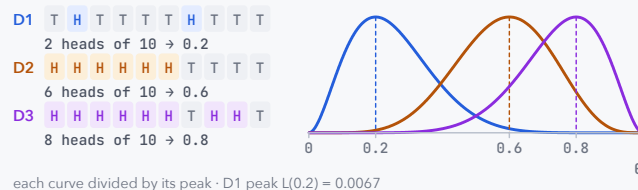


For i.i.d. data (the plate repeats x_n with shared μ, σ):

$$\mathcal{L}(\theta) = \prod_i f(x_i | \theta), \quad \log \mathcal{L}(\theta) = \sum_i \log f(x_i | \theta)$$

Maximising $\mathcal{L}$, maximising $\log \mathcal{L}$ and minimising the negative log-likelihood (NLL) give the same $\hat{\theta}$.

4 Bernoulli: the MLE is the fraction of heads



With $f(x | \theta) = \theta^x (1 - \theta)^{1-x}$, $x \in \{0, 1\}$:

$$\log \mathcal{L} = \sum_i [x_i \log \theta + (1 - x_i) \log(1 - \theta)]$$

$$\frac{d \log \mathcal{L}}{d\theta} = \sum_i \frac{x_i - \theta}{\theta(1 - \theta)} = 0 \Rightarrow \hat{\theta} = \frac{1}{n} \sum_i x_i$$

D1 has 2 heads in 10: $\hat{\theta} = 0.2$, $\mathcal{L}(0.2) = 0.2^2 \cdot 0.8^8 = 0.0067$.

5 Normal: the sample mean and the $1/n$ variance

$$\log \mathcal{L}(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - n \log \sigma - \frac{1}{2\sigma^2} \sum_i (x_i - \mu)^2$$

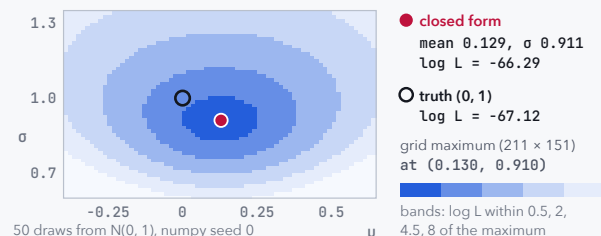
Set the derivatives in μ and in σ to zero:

$$\frac{1}{\sigma^2} \sum_i (x_i - \mu) = 0 \Rightarrow \hat{\mu} = \frac{1}{n} \sum_i x_i = \bar{x}$$

$$-\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_i (x_i - \mu)^2 = 0 \Rightarrow \hat{\sigma}^2 = \frac{1}{n} \sum_i (x_i - \hat{\mu})^2$$

$\hat{\mu}$ does not depend on σ , so plug it into $\hat{\sigma}^2$ for the joint maximum.

6 The closed form sits at the top of the surface

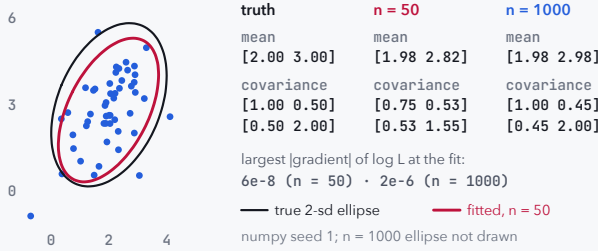


50 draws: $\bar{x} = 0.129$, $\hat{\sigma} = 0.911$, matched by a grid search. The truth (0, 1) scores lower (-67.12 against -66.29): on its own sample the MLE always wins. The lecture's run gave mean -0.0089 , SD 1.0376 (quoted).

Unbiased mean, biased variance

The same recipe for a multivariate normal, then what "on average" means for an estimator, and why dividing by n comes out too small.

7 Multivariate normal: the mean is still the average

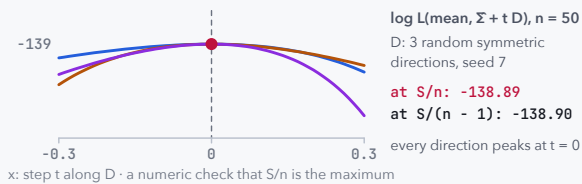


With $d_i = x_i - \mu$ and dimension k :

$$\log \mathcal{L}(\mu, \Sigma) = -\frac{nk}{2} \log 2\pi - \frac{n}{2} \log |\Sigma| - \frac{1}{2} \sum_i d_i^T \Sigma^{-1} d_i$$

Using $\nabla_a a^T A a = 2Aa$ for symmetric A (see the probability refresher sheet), $\nabla_\mu \log \mathcal{L} = \Sigma^{-1} \sum_i d_i = 0$, so $\hat{\mu} = \bar{x}$.

8 Covariance: the $1/n$ scatter matrix



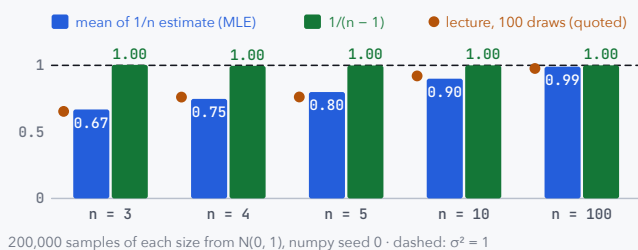
Write the precision $\Lambda = \Sigma^{-1}$ and the scatter $S = \sum_i (x_i - \hat{\mu})(x_i - \hat{\mu})^T$. As $a^T \Lambda a = \text{tr}(\Lambda a a^T)$:

$$\log \mathcal{L} = \text{const} + \frac{n}{2} \log |\Lambda| - \frac{1}{2} \text{tr}(\Lambda S)$$

$$\frac{n}{2} \Lambda^{-1} - \frac{1}{2} S = 0 \Rightarrow \hat{\Sigma} = \frac{1}{n} S$$

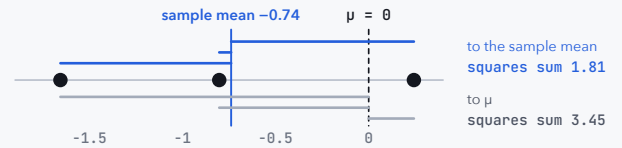
from $\partial_\Lambda \log |\Lambda| = \Lambda^{-1}$ and $\partial_\Lambda \text{tr}(\Lambda S) = S$. By invariance (panel 11) the maximiser in Λ gives the one in Σ . In numpy: `np.cov(X.T, bias=True)`.

9 Bias: the average error over repeated datasets



Bias($\hat{\theta}$) = $\mathbb{E}[\hat{\theta}] - \theta$, the expectation taken over datasets drawn from the model. $\mathbb{E}[\hat{\mu}] = \frac{1}{n} \sum_i \mathbb{E}[X_i] = \mu$, for any distribution with mean μ . $\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n} \sigma^2$: two thirds of the truth at $n = 3$, 1% short at $n = 100$.

10 The sample mean is fitted to the same points



$3.45 = 1.81 + 3 \times (-0.74)^2$: the gap is $n(\text{mean} - \mu)^2$
 three draws from $N(0, 1)$, numpy seed 9

$\sum (x_i - \mu)^2 = \sum (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2$, so squares about $\bar{x}$ never exceed those about μ . With $\mathbb{E}[X^2] = \sigma^2 + \mu^2$ and $\mathbb{E}[\bar{X}^2] = \sigma^2/n + \mu^2$:

$$\mathbb{E}[\hat{\sigma}^2] = \mathbb{E}\left[\frac{1}{n} \sum X_i^2\right] - \mathbb{E}[\bar{X}^2] = \frac{n-1}{n} \sigma^2$$

With μ known, $\frac{1}{n} \sum (x_i - \mu)^2$ is unbiased. Otherwise multiply by $n/(n-1)$ (Bessel's correction):

```
np.var(x)           # 1/n (ddof=0): MLE
np.var(x, ddof=1)   # 1/(n-1)
torch.var(x)         # 1/(n-1) by default
torch.var(x, correction=0) # 1/n
pd.Series(x).var()   # 1/(n-1)
```

11 Invariance: transform the MLE, not the bias



If $\hat{\theta}$ is the MLE of θ , then $g(\hat{\theta})$ is the MLE of $g(\theta)$: $\hat{\sigma} = \sqrt{\hat{\sigma}^2}$, and the odds for D1 are $0.2/0.8 = 0.25$. So solving for σ or σ^2 gives one answer. Unbiasedness does not pass through a nonlinear g : s^2 is unbiased, yet $\mathbb{E}[s] = 0.886\sigma$ at $n = 3$.

12 Pause and think

Three tosses, all heads: the MLE? $\hat{\theta} = 1$, so tails "never" happen. Little data gives an overconfident fit; a prior helps.

One point from a normal: $\hat{\sigma}^2$? Zero; $\log \mathcal{L}$ grows without bound as $\sigma \rightarrow 0$.

Why maximise $\log \mathcal{L}$, not $\mathcal{L}$? Same maximiser; sums are easier to differentiate and do not underflow.

Does $\mathbb{E}[\hat{\mu}] = \mu$ need normal data? No. Linearity of expectation only needs $\mathbb{E}[X_i] = \mu$.

maximise $\log p(D | \theta)$ over θ · normal MLE: mean and $1/n$ covariance · the $1/n$ variance is low by $(n-1)/n$

Source: lecture slides "Maximum likelihood estimation" and the notebooks on univariate, multivariate and Bernoulli MLE and the biased variance estimate, nipunbatra.github.io/pml-teaching. Numbers recomputed in NumPy with fixed seeds. Fixed here: the log-likelihood has $\log |\Sigma|$, not $\log \Sigma$; $(x - \mu)^T \Sigma^{-1}$ is the transpose of $\Sigma^{-1}(x - \mu)$, not equal to it; the slides' Bernoulli likelihood curves are e times too tall.